

11.1.3 根据定义判断下列级数是否收敛,若收敛求出其和:

(1) $\sum_{n=1}^{\infty} (\sqrt{n+1} - \sqrt{n})$;

(2) $\frac{1}{1 \cdot 3} + \frac{1}{3 \cdot 5} + \frac{1}{5 \cdot 7} + \cdots + \frac{1}{(2n-1)(2n+1)} + \cdots$;

(3) $\frac{1}{2} + \frac{1}{3} + \frac{1}{2^2} + \frac{1}{3^2} + \cdots + \frac{1}{2^n} + \frac{1}{3^n} + \cdots$;

(4) $\sin \frac{\pi}{6} + \sin \frac{2}{6} \pi + \cdots + \sin \frac{n}{6} \pi + \cdots$.

11.1.4 用级数的性质判断下列级数的敛散性:

(1) $\frac{1}{3} + \frac{1}{\sqrt{3}} + \frac{1}{\sqrt[3]{3}} + \cdots + \frac{1}{\sqrt[n]{3}} + \cdots$;

(2) $\frac{1}{3} + \frac{1}{6} + \frac{1}{9} + \cdots + \frac{1}{3n} + \cdots$;

(3) $\left(\frac{1}{2} + \frac{1}{3}\right) + \left(\frac{1}{2^2} + \frac{1}{3^2}\right) + \cdots + \left(\frac{1}{2^n} + \frac{1}{3^n}\right) + \cdots$.

11.1.5 设有级数 $\langle A \rangle$: $a_1 + a_2 + a_3 + a_4 + \cdots + a_{2n-1} + a_{2n} + \cdots$ 收敛,和为 S .

求证:级数 $\langle A' \rangle$: $a_2 + a_1 + a_4 + a_3 + \cdots + a_{2n} + a_{2n-1} + \cdots$ 也收敛,和也是 S .

(1) $S_n = \sqrt{n+1} - 1 \rightarrow \infty$, 发散

(2) $\sum_{n=1}^n \left(\frac{1}{2n-1} - \frac{1}{2n+1} \right) \frac{1}{2} = \left(1 - \frac{1}{2n+1} \right) \cdot \frac{1}{2}$

$\lim_{n \rightarrow \infty} S_n = \frac{1}{2}$

(3) $\sum_{i=1}^n \left(\frac{1}{2^i} + \frac{1}{3^i} \right) = 1 - \frac{1}{2^n} + \frac{1}{2} \left(1 - \frac{1}{3^n} \right) = \frac{3}{2} - \frac{1}{2^n} - \frac{1}{2 \cdot 3^n}$

$\lim_{n \rightarrow \infty} S_n = \frac{3}{2}$

(4) $\sum_{k=1}^n \sin \frac{k}{6} \pi$

$\frac{1}{\sin \frac{\pi}{12}} \cdot \sum_{k=1}^n \sin \frac{k}{12} \pi \cdot \sin \frac{k}{6} \pi = -\frac{1}{2 \sin \frac{\pi}{12}} \cdot \sum_{k=1}^n \left(\cos \frac{2k+1}{12} \pi - \cos \frac{2k-1}{12} \pi \right)$
 $= -\frac{1}{2 \sin \frac{\pi}{12}} \left(\cos \frac{(2n+1)\pi}{12} - \cos \frac{\pi}{12} \right)$

而 $\lim_{n \rightarrow \infty} S_n$ 不存在, 故级数发散

设 A, A' 的前 n 项和分别为 S_n, T_n

则必有 $S_{2n} = T_{2n}$,

记 $\lim_{n \rightarrow \infty} S_{2n} = S$, 则 $\lim_{n \rightarrow \infty} T_{2n} = S$

而 $\lim_{n \rightarrow \infty} T_{2n+1} = \lim_{n \rightarrow \infty} (T_{2n} + a_{2n+1}) = S$

则 $\lim_{n \rightarrow \infty} T_n = S$

故 A' 也收敛到 S

11.2.2 利用级数的性质判断下列级数的敛散性:

(1) $\frac{1}{1} + \frac{1}{\sqrt{2}} + \frac{1}{\sqrt[3]{3}} + \frac{1}{\sqrt[4]{4}} + \cdots$; $\frac{1}{n^k}$ (1) $\sum_{n=1}^{\infty} \frac{1}{n^k}$, $\lim_{n \rightarrow \infty} u_n = 1$ 发散

(2) $1 + \frac{2}{3} + \frac{3}{5} + \frac{4}{7} + \frac{5}{9} + \cdots$;

(3) $\left(1 + \frac{2}{3}\right) + \left(\frac{1}{2} + \frac{2^2}{3^2}\right) + \left(\frac{1}{3} + \frac{2^3}{3^3}\right) + \cdots$; (2) $\sum_{n=1}^{\infty} \frac{n}{2n-1}$ $\lim_{n \rightarrow \infty} u_n = \frac{1}{2}$ 发散

(3) 发散 + 收敛 发散

(4) $\frac{1}{4^2} + \frac{1}{5^2} + \frac{1}{6^2} + \frac{1}{7^2} + \cdots$; 收敛

(5) $\frac{1}{5} + \frac{1}{10} + \frac{1}{15} + \cdots + \frac{1}{5n} + \cdots$. 发散

11.2.3 利用比值或根值判别法判断下列正项级数的敛散性:

(1) $\sum_{n=1}^{\infty} \frac{3^n n!}{n^n}$; (2) $\sum_{n=1}^{\infty} n \left(\frac{3}{4}\right)^n$; (3) $\sum_{n=1}^{\infty} \frac{n^4}{n!}$;
(4) $\sum_{n=1}^{\infty} \frac{1 \cdot 3 \cdot 5 \cdots (2n-1)}{1 \cdot 4 \cdot 7 \cdots (3n-2)}$; (5) $\sum_{n=1}^{\infty} \frac{3^{2n-1}}{(2n-1)2^{2n-1}}$.

11.2.4 找收敛的强级数或发散的弱级数判断下列正项级数的敛散性:

(1) $\sum_{n=1}^{\infty} \frac{3}{n(n+1)}$; (2) $\sum_{n=1}^{\infty} \frac{2}{n^n}$;
(3) $\sum_{n=1}^{\infty} \frac{1}{3^n+1}$; (4) $\sum_{n=2}^{\infty} \frac{\ln n}{n^{2/3}}$.

(3) $u_n = \frac{n^4}{n!}$, $\frac{u_n}{u_{n+1}} = \frac{1}{n} \left(\frac{n}{n+1}\right)^4$,

$\lim_{n \rightarrow \infty} \frac{u_n}{u_{n+1}} = 0$, 收敛

(5). $\frac{u_n}{u_{n+1}} = \frac{9}{4} \cdot \frac{2n-3}{2n-1} \rightarrow \frac{9}{4}$, 故发散.

(1) $\frac{u_n}{u_{n-1}} = \frac{3n}{n^n} \cdot (n-1)^{n-1} = \left(\frac{n-1}{n}\right)^n \cdot \frac{3n}{n-1}$

$\lim_{n \rightarrow \infty} \frac{u_n}{u_{n-1}} = 3 \lim_{n \rightarrow \infty} e^{n \ln(1-\frac{1}{n})} = \frac{3}{e} > 1$, 故发散

(2) $\lim_{n \rightarrow \infty} n \sqrt[n]{u_n} = \frac{3}{4} < 1$, 收敛

(4) $u_n = \frac{(2n-1)!}{(3n-2)!}$, $\frac{u_n}{u_{n+1}} = \frac{2n-1}{3n-2} \rightarrow \frac{2}{3}$, 收敛

(5)

(1) $2n+1 > 3$, 则 $\frac{\frac{3}{n(n+1)}}{\frac{1}{n^2}} = \frac{3n^2}{n^2+n} \rightarrow 3$, 故收敛

$n+1 > n$, $n(n+1) > n^2$.

(2) 当 $n > 2$ 时, 有 $n^n > n^2$, $\therefore \frac{2}{n^n} < \frac{2}{n^2}$, 收敛

(3) $\frac{1}{3^n+1} < \frac{1}{3^n}$ 收敛

(4) $\frac{\ln n}{n^{\frac{1}{2}}} > \frac{\ln n}{n^2}$ $\ln n > 1$, $\frac{\ln n}{n^{\frac{1}{2}}} > \frac{1}{n^{\frac{1}{2}}}$

$\ln n \leq n-1$

11.2.5 用比较原理的极限形式判断下列正项级数的敛散性:

(1) $\sum_{n=1}^{\infty} \frac{1}{3^n-n}$; (2) $\sum_{n=1}^{\infty} \frac{2n+1}{(n+1)(n+2)(n+3)}$;
(3) $\sum_{n=1}^{\infty} \frac{4n}{(n+1)(n+2)}$; (4) $\sum_{n=1}^{\infty} 2^n \sin \frac{\pi}{3^n}$.

(1) $\lim_{n \rightarrow \infty} \frac{\frac{1}{3^n-n}}{\frac{1}{n^2}} = 0$. 故收敛

(3) $\frac{\frac{4n}{(n+1)(n+2)}}{\frac{1}{n}} = 4$, 发散

(2) $\lim_{n \rightarrow \infty} \frac{\frac{2n+1}{(n+1)(n+2)(n+3)}}{\frac{1}{n^2}} = 2$, 故收敛

(4) $\frac{2^n \sin \frac{\pi}{3^n}}{\frac{2^n \pi}{3^n}} = 1$, 故收敛

(1) $\lim_{n \rightarrow \infty} \frac{\frac{1}{3^n-n}}{\frac{1}{3^n}} = 1$

11.2.6 确定一般项关于无穷小 $\frac{1}{n}$ 的阶来判断下列正项级数的敛散性:

(1) $\sum_{n=1}^{\infty} \left(\frac{1}{n} - \ln \left(1 + \frac{1}{n} \right) \right)$; (2) $\sum_{n=1}^{\infty} \left(1 - \cos \frac{1}{n} \right)^p$ ($p > 0$ 为常数).

(1) $\lim_{n \rightarrow \infty} \frac{\frac{1}{n} - \ln \left(1 + \frac{1}{n} \right)}{\frac{1}{n^2}} = 0$, $\lim_{n \rightarrow \infty} \frac{\frac{1}{n} - \ln \left(1 + \frac{1}{n} \right)}{\frac{1}{n^2}} = 0$, 故收敛.
 $\ln \left(1 + \frac{1}{n} \right)$ 并非独立在一个项式中, 不能直接等价无穷小!
 应该用洛必达法则

$\ln \left(1 + \frac{1}{n} \right) = \frac{1}{n} - \frac{1}{2n^2} + o \left(\frac{1}{n^2} \right)$

(2) $\lim_{n \rightarrow \infty} \frac{\left(1 - \cos \frac{1}{n} \right)^p}{\frac{1}{n^p}} = \lim_{n \rightarrow \infty} \left(n - n \cos \frac{1}{n} \right)^p = 0$

$1 - \cos \frac{1}{n} = 1 - \left(1 - \frac{1}{2n^2} + o \left(\frac{1}{n^2} \right) \right)$

$= \frac{1}{2n^2} + o \left(\frac{1}{n^2} \right)$

$\therefore \left(1 - \cos \frac{1}{n} \right)^p = \left(\frac{1}{2n^2} + o \left(\frac{1}{n^2} \right) \right)^p$

$1 - \cos \frac{1}{n} \sim \frac{1}{2n^2}$ 等价无穷小

11.2.7 判断下列正项级数的敛散性:

(1) $\sum_{n=1}^{\infty} \frac{n \cos^2 \frac{n\pi}{3}}{2^n}$; (2) $\sum_{n=1}^{\infty} \frac{n^{n-1}}{(2n^2 + n + 1)^{\frac{n+1}{2}}}$

(3) $\sum_{n=1}^{\infty} n \tan \frac{\pi}{2^{n+2}}$; (4) $\sum_{n=1}^{\infty} \frac{1}{na+b}$ ($a > 0, b > 0$ 为常数);

(5) $\sum_{n=1}^{\infty} \int_0^{\frac{\pi}{n}} \frac{\sin x}{1+x} dx$; * (6) $\sum_{n=1}^{\infty} \left(1 - \frac{\ln n}{n} \right)^n$.

* 11.2.8 用积分判别法讨论级数

$\sum_{n=3}^{\infty} \frac{1}{n(\ln n)^p (\ln \ln n)^q}$ ($p, q > 0$)

的敛散性.

(1) $\frac{n \cos^2 \frac{n\pi}{3}}{2^n} \leq \frac{n}{2^n}$, 收敛

(2) $\frac{n^{n-1}}{(2n^2 + n + 1)^{\frac{n+1}{2}}} < \frac{n^{n-1}}{(2n^2)^{\frac{n+1}{2}}} = \frac{n^{n-1}}{2^{\frac{n+1}{2}} \cdot n^{n+1}} = \frac{1}{(\sqrt{2})^{n+1} n^2} < \frac{1}{n^2}$, 收敛

(3) $\lim_{n \rightarrow \infty} \frac{n \cdot \tan \frac{\pi}{2^{n+2}}}{\frac{1}{2^{n+2}}} = 1$, 收敛

(4) $\sum_{n=1}^{\infty} \frac{1}{an+b}$ $\lim_{n \rightarrow \infty} \frac{\frac{1}{an+b}}{\frac{1}{n}} = \frac{1}{a}$, 发散

(5) $\int_0^{\frac{2}{n}} \frac{\sin x}{1+x} dx$ $\because \frac{2}{n} \leq \pi$, 在一个正区间内,

$\int_0^{\frac{2}{n}} \frac{\sin x}{1+x} dx \leq \int_0^{\frac{2}{n}} \sin x dx = -\cos x \Big|_0^{\frac{2}{n}} = 1 - \cos \frac{2}{n} \sim \frac{1}{2} \frac{2^2}{n^2}$, 故收敛

$$(6) \quad f(n) = \left(1 - \frac{\ln n}{n}\right)^n = e^{n \ln \left(1 - \frac{\ln n}{n}\right)},$$

$$\lim_{n \rightarrow \infty} \frac{e^{n \ln \left(1 - \frac{\ln n}{n}\right)}}{\frac{1}{n}} = \lim_{n \rightarrow \infty} \frac{e^{n \cdot \left(-\frac{\ln n}{n}\right)}}{\frac{1}{n}} = 1.$$

11.2.9 判断下列级数是否收敛, 绝对收敛还是条件收敛:

(1) $1 - \frac{1}{3} + \dots + (-1)^{n+1} \frac{1}{2n-1} + \dots;$

(2) $\frac{1}{\ln 2} - \frac{1}{\ln 3} + \frac{1}{\ln 4} - \frac{1}{\ln 5} + \dots;$

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(1) $u_n = \frac{1}{2n-1}$, 单减且 $\rightarrow 0$ 条件收敛

(2) $u_n = \frac{1}{\ln n}$, $\rightarrow 0$ 且单减 $\ln n < n$, $\frac{1}{\ln n} > \frac{1}{n}$ 条件收敛

(3) $\sum_{n=1}^{\infty} (-1)^{n+1} \frac{2^{n^2}}{n!};$ (4) $\sum_{n=1}^{\infty} (-1)^{\frac{n(n-1)}{2}} \frac{n^{10}}{2^n};$

(5) $\frac{1}{\pi^2} \sin \frac{\pi}{2} - \frac{1}{\pi^3} \sin \frac{\pi}{3} + \frac{1}{\pi^4} \sin \frac{\pi}{4} - \dots;$

(6) $\sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n^p};$ (7) $\sum_{n=1}^{\infty} \frac{(-1)^n}{n - \ln n};$

(8) $\sum_{n=2}^{\infty} \frac{(-1)^n \sqrt{n}}{n-1}.$

(3) $u_n = \frac{2^{n^2}}{n!}$

$\frac{u_n}{u_{n-1}} = \frac{2^{2n-1}}{n} > 1, \quad \lim_{n \rightarrow \infty} \frac{u_n}{u_{n-1}} = \infty, \text{ 发散}$

$\frac{2^{n^2}}{n!}$ 单增, $\lim_{n \rightarrow \infty} \frac{2^{n^2}}{n!} \neq 0$
发散

(4) $u_{4n} = (-1)^{2n(4n-1)} \cdot \frac{(4n)^{10}}{2^{4n}} = \frac{(4n)^{10}}{2^{4n}}$

$u_{4n+1} = (-1)^{2n(4n+1)} \cdot \frac{(4n+1)^{10}}{2^{4n+1}} = \frac{(4n+1)^{10}}{2^{4n+1}}$

$u_{4n+2} = (-1)^{(2n+1)(4n+1)} \cdot \frac{(4n+2)^{10}}{2^{4n+2}} = -\frac{(4n+2)^{10}}{2^{4n+2}}$

绝对收敛

$$u_{4n+3} = (-1)^{(2n+1)(4n+3)} \cdot \frac{(4n+3)^{10}}{2^{4n+3}} = - \frac{(4n+3)^{10}}{2^{4n+3}}$$

将级数两两加括号组合 原式 = $\sum_{n=1}^{\infty} (-1)^n \left(\frac{n^{10}}{2^n} + \frac{(n+1)^{10}}{2^{n+1}} \right)$

(5) $|u_n| = \left| \frac{1}{2^n} \cdot \sin \frac{2}{n} \right| \leq \frac{1}{2^n}$ 绝对收敛

(6) $p > 1$ 绝对收敛. $0 < p < 1$ 时条件收敛 $p \leq 0$ 时发散

(7) $|u_n| = \frac{1}{n - \ln n}$, $\frac{u_n}{\frac{1}{n}} = \frac{1}{1 - \frac{\ln n}{n}} \cdot \lim_{n \rightarrow \infty} = 1$, 发散

$\frac{1}{n - \ln n} \downarrow$ $\lim_{n \rightarrow \infty} \frac{1}{n - \ln n} = 0$, 条件收敛

(8) $u_n = \frac{\sqrt{n}}{n-1}$, $\downarrow \rightarrow 0$ 条件收敛

11.2.10 设级数 $\sum_{n=1}^{\infty} a_n, \sum_{n=1}^{\infty} b_n$ 均收敛, 且有
 $a_n \leq u_n \leq b_n \quad (n = 1, 2, \dots)$.

求证: $\sum_{n=1}^{\infty} u_n$ 收敛.

11.2.11 设正项级数 $\sum_{n=1}^{\infty} u_n$ 收敛, 求证:

(1) $\sum_{n=1}^{\infty} \sqrt{u_n u_{n+1}}$ 收敛; (2) $\sum_{n=1}^{\infty} \frac{u_n}{1-u_n}$ 收敛 ($u_n \neq 1$).

$\lim_{n \rightarrow \infty} u_n = 0 \Rightarrow u_n$ 有界 $\Rightarrow 1 - u_n$ 有下界

* 11.2.12 设 $\sum_{n=1}^{\infty} \frac{a_n}{n^{x_0}}$ 收敛, 求证: 当 $x > x_0$ 时 $\sum_{n=1}^{\infty} \frac{a_n}{n^x}$ 收敛.

(1) $u_n + u_{n+1} \geq 2\sqrt{u_n u_{n+1}}$

$\therefore \sum_{n=1}^{\infty} (u_n + u_{n+1}) \geq 2 \sum_{n=1}^{\infty} \sqrt{u_n u_{n+1}}$ 收敛

(2) $1 - \frac{1-u_{n+1}}{1-u_n} = -1 + \frac{1}{1-u_n}$ $1-u_n$ 是有界的

不能死板， a_n 的部分和有界，取 a_n 部分和

例 8. 设序列 $\{a_n\}$ 和 $\{b_n\}$ 分别满足如下性质： $\{a_n\}$ 的部分和 $S_n = \sum_{k=1}^n a_k$ 有界，正项级

数 $\sum_{n=1}^{\infty} |b_n - b_{n+1}|$ 收敛且 $\lim_{n \rightarrow \infty} b_n = 0$ 。证明级数 $\sum_{n=1}^{\infty} a_n b_n$ 收敛。

$$\begin{aligned} \sum_{n=n+1}^{n+p} a_n b_n &= a_{n+1} b_{n+1} + a_{n+2} b_{n+2} + \dots + a_{n+p} b_{n+p} \\ &= a_{n+1} (B_{n+1} - B_n) + a_{n+2} (B_{n+2} - B_{n+1}) + \dots + a_{n+p} (B_{n+p} - B_{n+p-1}) \\ &= (a_{n+1} - a_{n+2}) B_{n+1} + (a_{n+2} - a_{n+3}) B_{n+2} + \dots + a_{n+p} B_{n+p} - a_{n+1} B_{n+1} \\ &= \left[\sum_{n=n+1}^{n+p-1} (a_n - a_{n+1}) \cdot B_n \right] + a_{n+p} B_{n+p} - a_{n+1} B_{n+1} \end{aligned}$$

$$\begin{aligned} \sum_{n=1}^{\infty} |b_n - b_{n+1}| \text{ 收敛, } \sum_{n=1}^n |b_n - b_{n+1}| &= |b_1 - b_2| + |b_2 - b_3| + \dots + |b_n - b_{n+1}| < B_{n+1} + B_n - B_1 \\ &= |B_1 - (B_2 - B_1)| + |B_2 - B_1 - (B_3 - B_2)| + \dots + |B_n - B_{n-1} - (B_{n+1} - B_n)| \\ &= |2B_1 - B_2| + |2B_2 - B_1 - B_3| + \dots + |2B_n - B_{n+1} - B_{n-1}| \end{aligned}$$

$$\begin{aligned} \sum_{n=n+1}^{n+p} a_n b_n &= \sum_{n=n+1}^{n+p-1} A_n (b_n - b_{n+1}) + A_{n+p} b_{n+p} - A_{n+1} b_{n+1} \\ \left| \sum_{n=n+1}^{n+p} a_n b_n \right| &\leq \left| \sum_{n=n+1}^{n+p-1} A_n (b_n - b_{n+1}) \right| + |A_{n+p}| \cdot |b_{n+p}| + |A_{n+1}| \cdot |b_{n+1}| \\ &\leq M \cdot \sum_{n=n+1}^{n+p} |b_n - b_{n+1}| + M \cdot |b_{n+p}| + M \cdot |b_{n+1}| \end{aligned}$$

$$\because \sum_{n=1}^{\infty} |b_n - b_{n+1}| \text{ 收敛, 由柯西准则 } \forall \varepsilon > 0, \exists N, n > N, \sum_{k=n}^{n+p} |b_k - b_{k+1}| < \frac{\varepsilon}{3M}$$

$$\text{而 } \lim_{n \rightarrow \infty} b_n = 0, \forall \varepsilon > 0, \exists N_2, n > N_2, |b_n| < \frac{\varepsilon}{3M}$$

$$\therefore \left| \sum_{n=n+1}^{n+p} a_n b_n \right| \leq \varepsilon$$

2.1 例题

题 1. 设参数 p 是正实数, 判断正项级数收敛性 $\sum_{n=1}^{\infty} \frac{1}{p^{\ln n}}$ 。

$$p^{\ln n} = e^{(\ln n) \cdot \ln p} = n^{\ln p}$$

$$\frac{1}{p^{\ln n}} = \frac{1}{n^{\ln p}}$$

$$\begin{cases} p > e, \text{收敛} \\ p \leq e, \text{发散} \end{cases}$$

题 2. 设参数 p 和 q 都是正实数, 判断正项级数收敛性

1. $\sum_{n=1}^{\infty} \frac{(\ln n)^p}{n^q}$ 。

2. $\sum_{n=2}^{\infty} \frac{1}{n^q (\ln n)^p}$ 。

3. $\sum_{n=2}^{\infty} \frac{1}{n^q (\ln \ln n)^p}$ 。

1. ① $0 < q \leq 1$, $\frac{(\ln n)^p}{n^q} \geq \frac{1}{n^q}$, 发散

② $q > 1$, $(\ln n)^p < n^{\epsilon q}$, $q > s > 1$.

$$\lim_{n \rightarrow \infty} \frac{\frac{(\ln n)^p}{n^q}}{\frac{1}{n^s}} = \lim_{n \rightarrow \infty} \frac{(\ln n)^p}{n^{q-s}} = 0,$$

题 3. 设 $\sum_{n=1}^{\infty} a_n$ 是一个 正项级数，其部分和序列 $S_n = \sum_{k=1}^n a_k$ ，证明下述命题：

1. 如果 $\sum_{n=1}^{\infty} a_n$ 发散，那么正项级数 $\sum_{n=1}^{\infty} \frac{a_n}{S_n}$ 也发散。

2. 不论 $\sum_{n=1}^{\infty} a_n$ 是否收敛，正项级数 $\sum_{n=1}^{\infty} \frac{a_n}{S_n^2}$ 总收敛。

分析：本题是抽象级数的证明题，整体思路是采用比较判别法或柯西准则，对正

1. 若 $\sum_{n=1}^{\infty} \frac{a_n}{S_n}$ 收敛，即

$$\forall \varepsilon > 0, \exists N, \left| \sum_{n=n+1}^{n+p} \frac{a_n}{S_n} \right| \leq \frac{\sum_{n=n+1}^{n+p} a_n}{S_{n+1}} = \frac{S_{n+p} - S_n}{S_{n+1}} \leq \frac{S_{n+p}}{S_n} - 1 \leq \varepsilon \quad S_{n+p} \leq (\varepsilon + 1) S_n,$$

$$p=1, S_{n+1} \leq (\varepsilon + 1) S_n \leq (\varepsilon + 1)^2 S_{n-1} \leq \dots \leq (\varepsilon + 1)^n a_1$$

故 S_n 有界，而 $S_n \uparrow$ ，故 $\lim_{n \rightarrow \infty} S_n$ 存在

即 $\sum_{n=1}^{\infty} a_n$ 收敛，矛盾

故 $\sum_{n=1}^{\infty} \frac{a_n}{S_n}$ 发散

$$2. \quad \forall \varepsilon > 0, p > 1 \quad \left| \sum_{n=n+1}^{n+p} \frac{a_n}{S_n^2} \right| \leq \sum_{n=n+1}^{n+p} \frac{S_n - S_{n-1}}{S_n S_{n-1}} = \sum_{n=n+1}^{n+p} \left(\frac{1}{S_{n-1}} - \frac{1}{S_n} \right) = \frac{1}{S_n} - \frac{1}{S_{n+p}} = \frac{\sum_{n=n+1}^{n+p} a_n}{S_n S_{n+p}} \leq \frac{1}{S_n} \leq \frac{1}{a_1}$$

$$\frac{a_n}{S_n^p} (p > 1) \leq \frac{a_n}{S_n^{p-1} S_{n+1}} = \frac{S_n - S_{n-1}}{S_n^{p-1} S_{n+1}} = \frac{1}{S_n^{p-2} S_{n+1}} - \frac{1}{S_n^{p-1}}$$

1. 求下列幂级数的收敛区间及收敛域:

(1) $\sum_{n=1}^{\infty} \frac{x^n}{n2^n}$; (2) $\sum_{n=1}^{\infty} \frac{3^n + (-2)^n}{n} (x+1)^n$;

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(3) $x + \frac{1}{2} \cdot \frac{x^3}{3} + \frac{1 \cdot 3}{2 \cdot 4} \cdot \frac{x^5}{5} + \frac{1 \cdot 3 \cdot 5}{2 \cdot 4 \cdot 6} \cdot \frac{x^7}{7} + \dots$;

(4) $\sum_{n=0}^{\infty} 4^{n^2} x^{n^2}$; (5) $\sum_{n=1}^{\infty} (-1)^{n+1} \frac{(3n-1)^{3n} (x-2)^n}{(2n-3)^{3n}}$;

(6) $\sum_{n=1}^{\infty} \left(1 + \frac{1}{n}\right)^{n^2} (x-1)^n$.

(4). $\left| \frac{u_{n+1}}{u_n} \right| = 4^{2n+1} |x|^{2n+1} < 1$.

$\therefore |x|^{2n+1} < \frac{1}{4^{2n+1}}$

$\therefore |x| < \frac{1}{4}$, $R = \frac{1}{4}$, 区间: $(-\frac{1}{4}, \frac{1}{4})$

在 $\frac{1}{4}$ 时, 易知发散, 故收敛域: $(-\frac{1}{4}, \frac{1}{4})$

(5). $n\sqrt[n]{|u_n|} = \left(\frac{3n-1}{2n-3}\right)^3$, $\lim_{n \rightarrow \infty} n\sqrt[n]{|u_n|} = \frac{27}{8}$, $\therefore R = \frac{8}{27}$.

(6) $\lim_{n \rightarrow \infty} n\sqrt[n]{|u_n|} = e$, $R = \frac{1}{e}$,

(2). $a_n = \frac{3^n}{n} \left(1 + \left(-\frac{2}{3}\right)^n\right)$, $\lim_{n \rightarrow \infty} n\sqrt[n]{a_n} = 3$, $R = \frac{1}{3}$

$\therefore$ 区间: $(-\frac{4}{3}, \frac{2}{3})$

① $x = \frac{2}{3}$ 时, $a_n = \frac{1}{n} \cdot \left(1 + \left(-\frac{2}{3}\right)^n\right) > \frac{1}{3n}$, 发散.

② $x = -\frac{4}{3}$ 时, $a_n = \frac{1}{n} \left[(-1)^n + \left(\frac{2}{3}\right)^n\right] = \frac{(-1)^n}{n} + \frac{\left(\frac{2}{3}\right)^n}{n}$, 而 $\frac{(-1)^n}{n}$ 收敛; $\lim_{n \rightarrow \infty} n\sqrt[n]{\left|\frac{2}{3}\right|^n} = \frac{2}{3} < 1$, 收敛,

收敛域: $\left[-\frac{4}{3}, -\frac{2}{3}\right)$

(3) $u_n = \frac{(2n-1)!!}{(2n)!!} \frac{x^{2n+1}}{2n+1}$

$\left| \frac{u_{n+1}}{u_n} \right| = |x|^2 \cdot \frac{2n+1}{2n+2} \cdot \frac{2n+3}{2n+1}$, $\lim_{n \rightarrow \infty} \left| \frac{u_{n+1}}{u_n} \right| = |x|^2 < 1$, 故收敛区间: $(-1, 1)$

$x=1$ 时, $a_n = \frac{(2n-1)!!}{(2n)!!} \frac{1}{2n+1} < \frac{1}{\sqrt{2n+1}} \frac{1}{2n+1} = \frac{1}{(2n+1)^{\frac{3}{2}}}$, 收敛

$x=-1$ 时, $a_n = -\frac{(2n-1)!!}{(2n)!!} \frac{1}{2n+1}$, $n\left(\frac{u_n}{u_{n+1}} - 1\right) = \left(\frac{2n+2}{2n+1} \cdot \frac{2n+3}{2n+1} - 1\right) \cdot n = \left(\frac{4n^2+10n+6}{4n^2+4n+1} - 1\right) n = \frac{(6n+5)n}{4n^2+4n+1}$

$\lim_{n \rightarrow \infty} n\left(\frac{u_n}{u_{n+1}} - 1\right) = \frac{3}{2} > 1$

故收敛

$\therefore$ 收敛域: $[-1, 1]$

11.3.1 求下列级数的收敛区间及收敛域:

(1) $1 + \frac{x}{a} + \frac{x^2}{2a^2} + \frac{x^3}{3a^3} + \dots \quad (a > 0);$

(2) $x - \frac{x^3}{3 \cdot 3!} + \frac{x^5}{5 \cdot 5!} - \dots;$

(3) $\ln x + (\ln x)^2 + (\ln x)^3 + \dots;$

(4) $\sum_{n=0}^{\infty} \frac{1}{2n+1} \left(\frac{1-x}{1+x} \right)^n \quad \left(\text{提示: 令 } t = \frac{1-x}{1+x} \right);$

(5) $\sum_{n=1}^{\infty} 2^n x^{2n-1};$

(6) $\sum_{n=2}^{\infty} \frac{x^{n-1}}{n \cdot 3^n \cdot \ln n};$

(7) $\sum_{n=1}^{\infty} \frac{n! (x-2)^n}{n^n};$

(8) $\sum_{n=1}^{\infty} \left(1 + \frac{1}{2} + \dots + \frac{1}{n} \right) x^n;$

(9) $\sum_{n=1}^{\infty} \left(\frac{a^n}{n} + \frac{b^n}{n^2} \right) x^n, \quad a > 0, b > 0.$

(1) $U_n = \frac{x^n}{na^n}, \quad \lim_{n \rightarrow \infty} \frac{U_{n+1}}{U_n} = \lim_{n \rightarrow \infty} \frac{n}{n+1} \frac{1}{a} = \frac{1}{a}, \therefore R = a$

$x=a$, 发散, $x=-a$, 收敛 $[-a, a)$

(2) $\left| \frac{U_{n+1}}{U_n} \right| = \frac{(2n-1)(2n-1)!}{(2n+1)(2n+1)!} |x|^2 = \frac{2n-1}{2n \cdot (2n+1)^2} |x|^2 \quad \lim_{n \rightarrow \infty} \left| \frac{U_{n+1}}{U_n} \right| = 0, R = +\infty$

(3) $-1 < \ln x < 1, \quad \frac{1}{e} < x < e,$
 $x=e$ 时, 发散; $x=-\frac{1}{e}$ 时, 发散

(4) $-1 < \frac{1-x}{1+x} < 1, \quad \frac{1-x}{1+x} = -1, \quad 1-x = -1-x, \quad x$
 $\frac{1-x}{1+x} = -1 + \frac{2}{1+x} > -1, \quad \frac{2}{1+x} > 0, \quad x > -1$

$-1 + \frac{2}{1+x} < 1, \quad \frac{2}{1+x} < 2, \quad 1+x > 1, \quad x > 0$

(5) $\left| \frac{U_{n+1}}{U_n} \right| = 2 \cdot |x|^2 < 1, \quad |x|^2 < \frac{1}{2}, \quad R = \frac{1}{\sqrt{2}}$
 $x = \frac{1}{\sqrt{2}}, \quad \sqrt{2} \cdot 2^{\frac{1}{2}} \cdot 2^n = \sqrt{2}.$

(6) $\lim_{n \rightarrow \infty} \frac{U_{n+1}}{U_n} = \frac{1}{3} \lim_{n \rightarrow \infty} \frac{(n+1) \ln(n+1)}{n \cdot \ln n} = \frac{1}{3} \lim_{n \rightarrow \infty} \frac{\ln(n+1)+1}{\ln n+1} = \frac{1}{3} \lim_{n \rightarrow \infty} \frac{n+1}{n} = \frac{1}{3}$

$R=3,$ $\frac{1}{n \ln n} \quad f(n) = \frac{1}{n \ln n}, \quad \int_2^{+\infty} \frac{1}{n \ln n} dn = \lim_{A \rightarrow \infty} \int_2^A \frac{1}{\ln n} d \ln n = \lim_{A \rightarrow \infty} \ln \ln A = +\infty$

(7) $\lim_{n \rightarrow \infty} \frac{U_{n+1}}{U_n} = \lim_{n \rightarrow \infty} \frac{(n+1)n^n}{(n+1)^{n+1}} = \frac{1}{e} \quad R=e.$
 $(2e, 2+e)$

$x=2+e$ 时, $\frac{U_{n+1}}{U_n} = \left(\frac{n}{n+1} \right)^n e^n = \frac{e^n}{\left(1+\frac{1}{n} \right)^n} > 1$, 单增, $\therefore$ 发散

$x=2-e$ 时, 同理 $\frac{U_{n+1}}{U_n} = \frac{(-e)^n}{\left(1+\frac{1}{n} \right)^n} = (-1)^n \left(\frac{e}{1+\frac{1}{n}} \right)^n, \quad \left| \frac{U_{n+1}}{U_n} \right| > 1$, 发散

发散
 $\frac{(-1)^n}{n \ln n}, \quad \rightarrow 0, \downarrow, \text{收敛}$
 $[-3, 3)$

$$(8) \quad (-1, 1)$$

$$(4) \quad \sum_{n=1}^{\infty} \frac{a^n}{n} x^n + \sum_{n=1}^{\infty} \frac{b^n}{n^2} x^n$$

$$\lim_{n \rightarrow \infty} \sqrt[n]{\frac{a^n}{n}} = a, \quad \lim_{n \rightarrow \infty} \sqrt[n]{\frac{b^n}{n^2}} = b$$

$$R = \frac{1}{a}, \quad R = \frac{1}{b}$$

$$\therefore R = \min\left\{\frac{1}{a}, \frac{1}{b}\right\}$$

$$\textcircled{1} \text{ 若 } \frac{1}{a} < \frac{1}{b}, \quad R = \frac{1}{a}, \quad x = \frac{1}{a}, \quad \sum_{n=1}^{\infty} \left(\frac{1}{n} + \frac{b^n}{n^2 a^n}\right), \text{ 发散}; \quad x = -\frac{1}{a}, \text{ 收敛} \quad \left[-\frac{1}{a}, \frac{1}{a}\right)$$

$$\textcircled{2} \quad \frac{1}{a} > \frac{1}{b}, \quad R = \frac{1}{b}, \quad x = \frac{1}{b} \text{ 时 } \sum_{n=1}^{\infty} \left(\frac{a}{b}\right)^n \cdot \frac{1}{n} \text{ 收敛, 收敛} \quad x = -\frac{1}{b}, \quad (-1)^n \left(\frac{a}{b}\right)^n \cdot \frac{1}{n}, \text{ 收敛} \quad \left[-\frac{1}{b}, \frac{1}{b}\right]$$

$$a = b, \quad \text{收敛}$$

11.3.2 利用逐项求导与逐项积分, 求下列幂级数的和:

$$(1) \quad \sum_{n=1}^{\infty} \frac{(-1)^{n-1} x^{2n}}{n(2n-1)}, \quad |x| < 1; \quad (2) \quad \sum_{n=1}^{\infty} \frac{x^{4n+1}}{4n+1}, \quad |x| < 1;$$

$$(3) \quad \sum_{n=1}^{\infty} \frac{2n-1}{2^n} x^{2n-2}, \quad |x| \leq 1; \quad (4) \quad \sum_{n=1}^{\infty} (-1)^{n-1} \frac{x^n}{n}, \quad |x| < 1;$$

$$(5) \quad \sum_{n=0}^{\infty} (n+1)x^n, \quad |x| < 1; \quad (6) \quad \sum_{n=1}^{\infty} n(n+1)x^{n-1}, \quad |x| < 1.$$

$$(1) \quad f(x) = \sum_{n=1}^{\infty} \frac{(-1)^{n-1} x^{2n}}{n(2n-1)}, \quad |x| < 1, \quad f'(x) = \sum_{n=1}^{\infty} \frac{2(-1)^{n-1} x^{2n-1}}{2(2n-1)}, \quad f''(x) = \sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{2} x^{2n-2} = \sum_{n=0}^{\infty} \frac{(-1)^n}{2} x^{2n}$$

$$= \frac{1}{2} \cdot (1 - x^2 + x^4 - \dots + (-1)^n x^{2n} + \dots) = \frac{1}{2} \lim_{n \rightarrow \infty} \frac{1 - (-x^2)^{n+1}}{1 + x^2} = \frac{1}{2} \frac{1}{1+x^2}$$

$$f'(0) = 0 \quad \therefore f'(x) = \frac{1}{2} \arctan x$$

$$f(x) = \frac{1}{2} \int_0^x \arctan t \, dt = \frac{1}{2} \left(x \arctan x - \int_0^x \frac{t}{1+t^2} dt \right)$$

$$= \frac{1}{2} x \arctan x - \frac{1}{4} \ln(1+x^2) \quad 2x \arctan x - \ln(1+x^2)$$

$$(2) \quad f(x) = \sum_{n=1}^{\infty} \frac{x^{4n+1}}{4n+1}, \quad |x| < 1, \quad f'(x) = \sum_{n=1}^{\infty} x^{4n} = \lim_{n \rightarrow \infty} \frac{x^4(1-(x^4)^n)}{1-x^4} = \frac{x^4}{1-x^4}$$

$$\int_0^x \frac{t^4}{1-t^4} dt = \int_0^x \left(-1 + \frac{1}{1-t^4}\right) dt = -x + \frac{1}{2} \int_0^x \left(\frac{1}{1-t^2} + \frac{1}{1+t^2}\right) dt$$

$$= -x + \frac{1}{4} \left(\int_0^x \frac{1}{1-t} dt + \int_0^x \frac{1}{1+t} dt \right) + \frac{1}{2} \arctan x$$

$$= -x + \frac{1}{4} \cdot \ln \frac{1+x}{1-x} + \frac{1}{2} \arctan x$$

$$(3) \text{ 逐项积分 } \sum_{n=1}^{\infty} \frac{x^{2n-1}}{2^n} = \lim_{n \rightarrow \infty} \frac{\frac{x}{2} (1 - \frac{x^{2n}}{2^n})}{1 - \frac{x^2}{2}} = \frac{x}{2-x^2}, \quad f(x) = \left(\frac{x}{2-x^2} \right)' = \frac{2-x^2+2x^2}{(2-x^2)^2} = \frac{x^2+2}{(2-x^2)^2}$$

$$(4) \quad f(x) = \sum_{n=1}^{\infty} (-1)^{n+1} \frac{x^n}{n}, \quad f'(x) = \sum_{n=1}^{\infty} (-1)^{n+1} x^{n-1} = \sum_{n=0}^{\infty} (-1)^n x^n = \frac{1}{1+x}, \quad f(x) = \ln(1+x)$$

$$(5) \text{ 逐项积分, } \sum_{n=0}^{\infty} x^{n+1} = x + x^2 + \dots + x^n + \dots = \frac{x}{1-x} \quad (6) \text{ 积分两次, } \sum_{n=1}^{\infty} x^{n+1} = \frac{x^2}{1-x}$$

$$f(x) = \left(\frac{x}{1-x} \right)' = \frac{1-x+x}{(1-x)^2} = \frac{1}{(1-x)^2}$$

$$f(x) = \frac{2}{(1-x)^3}$$

11.3.3 利用某些幂级数的和函数,求下列级数的和:

$$(1) \sum_{n=1}^{\infty} \frac{2n-1}{2^n};$$

$$(2) \sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n \cdot (2n-1) \cdot 5^n};$$

$$(3) \sum_{n=0}^{\infty} \frac{n+1}{3^{n/2}};$$

$$(4) \sum_{n=1}^{\infty} \frac{n(n+1)}{2^n}.$$

$$(1) 3 \quad (2) 2 \cdot \frac{1}{\sqrt{5}} \arctan \frac{1}{\sqrt{5}} - \ln \frac{6}{5}$$

$$(3) \frac{1}{(1 - \frac{1}{\sqrt{3}})^2} = \frac{3}{(\sqrt{3}-1)^2} \quad (4) \sum_{n=1}^{\infty} \frac{n(n+1)}{2^{n-1}} = 16, \quad \sum_{n=1}^{\infty} \frac{n(n+1)}{2^n} = 8$$

11.4.1 求下列函数的马克劳林级数及其收敛区间:

- (1) $\frac{e^x - e^{-x}}{2}$; (2) $\ln(a+x)$; (3) $\sin^3 x$;
 (4) $\sin\left(\frac{\pi}{4} + x\right)$; (5) $\sqrt[3]{8-x^3}$; (6) $\arctan \frac{2x}{1-x^2}$;
 (7) $\arccos x$; (8) $\frac{3x-5}{x^2-4x+3}$; (9) $\frac{1}{\sqrt{4-x^2}}$;
 (10) $(1+x)e^{-x}$; (11) $\int_0^x \frac{\ln(1+x)}{x} dx$; (12) $\int_0^x \frac{dx}{\sqrt{1-x^4}}$.

$$(1) e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!}, e^{-x} = \sum_{n=0}^{\infty} \frac{(-1)^n x^n}{n!}$$

$$\therefore e^x - e^{-x} = \sum_{n=0}^{\infty} \frac{[1 - (-1)^n] x^n}{n!} = 2 \sum_{n=1}^{\infty} \frac{x^{2n-1}}{(2n-1)!}$$

$$\frac{e^x - e^{-x}}{2} = \sum_{n=1}^{\infty} \frac{x^{2n-1}}{(2n-1)!} \quad (-\infty + \infty)$$

$$(2) \ln a + \ln\left(1 + \frac{x}{a}\right)$$

$$= \ln a + \sum_{n=1}^{\infty} (-1)^{n+1} \frac{x^n}{n} \cdot \frac{1}{a^n} \quad [-a, a]$$

$$(3) \sin 3x = 3\sin x - 4\sin^3 x, \sin^3 x = \frac{3}{4}\sin x - \frac{1}{4}\sin 3x$$

$$\sin^3 x = \frac{3}{4} \sum_{n=1}^{\infty} (-1)^{n+1} \frac{x^{2n-1}}{(2n-1)!} - \frac{1}{4} \sum_{n=1}^{\infty} (-1)^{n+1} \frac{(3x)^{2n-1}}{(2n-1)!}$$

$$= \frac{1}{4} \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{(2n-1)!} \cdot (3 - 3^{2n-1}) x^{2n-1}$$

$$= \frac{1}{4} \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{(2n-1)!} 3(1 - 3^{2n-2}) x^{2n-1} \quad (n=1 \text{ 时 } 1-1=0)$$

$$= \frac{3}{4} \sum_{n=1}^{\infty} \frac{(-1)^n}{(2n)!} (1 - 3^{2n}) x^{2n+1}$$

$$(5) 2 \cdot 3 \sqrt{1 - \frac{x^3}{8}} = 2 \cdot \left(1 + \frac{1}{3} \left(-\frac{x^3}{8}\right) + \frac{1}{2} \cdot \frac{1}{3} \cdot \left(-\frac{2}{3}\right) \cdot \left(-\frac{x^3}{8}\right)^2 + \dots \right)$$

$$= 2 \cdot \left(1 + \sum_{n=1}^{\infty} \frac{(-1)^{2n+1}}{3^{n+1}} \frac{(2 \times 5 \times 8 \cdots (3n+2))}{(n+1)!} \cdot x^{3n+3} \right)$$

$$= 2 + 2 \cdot \sum_{n=1}^{\infty} \frac{(-1)^{2n+1}}{3^n n!} \cdot (2 \times 5 \times 8 \cdots (3n-1)) \left(\frac{x}{2}\right)^{3n}$$

$$(4) \sin\left(\frac{\pi}{4} + x\right) = \frac{\sqrt{2}}{2} (\sin x + \cos x)$$

$$= \frac{\sqrt{2}}{2} \left(\sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{(2n+1)!} + \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n}}{(2n)!} \right)$$

$$= \frac{\sqrt{2}}{2} \left(\sum_{n=0}^{\infty} (-1)^n \left(\frac{x^{2n}}{(2n)!} + \frac{x^{2n+1}}{(2n+1)!} \right) \right)$$

$$(6) 2 \arctan x = \arctan \frac{2x}{1-x^2}$$

$$\therefore \arctan \frac{2x}{1-x^2} = 2 \arctan x$$

$$(\arctan x)' = \frac{1}{1+x^2} = \sum_{n=0}^{\infty} (-1)^n x^{2n}, \arctan x = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{2n+1}$$

$$= 2 - \frac{x^3}{12} + 2 \cdot \sum_{n=2}^{\infty} \frac{(-1)^{n-3} \cdot (2 \times 5 \times \dots \times (3n-4))}{3^{n-1} (n-1)!} \left(\frac{x}{3}\right)^{3n-3}$$

$$\therefore \arctan \frac{2x}{1-x^2} = 2 \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{2n+1} \quad [x \in (-1, 1)]$$

$x = \pm 1$ 分母为 0

(7) $\arccos x$;

(8) $\frac{3x-5}{x^2-4x+3}$;

(9) $\frac{1}{\sqrt{4-x^2}}$;

(10) $(1+x)e^{-x}$;

(11) $\int_0^x \frac{\ln(1+x)}{x} dx$;

(12) $\int_0^x \frac{dx}{\sqrt{1-x^4}}$.

(7) $\arccos x = \frac{\pi}{2} - \arcsin x$

$$\begin{aligned} \text{而 } (\arcsin x)' &= \frac{1}{\sqrt{1-x^2}} = (1-x^2)^{-\frac{1}{2}} = 1 + \left(-\frac{1}{2}\right) \cdot (-x^2) + \frac{1}{2!} \left(-\frac{1}{2}\right) \left(-\frac{3}{2}\right) \cdot (-x^2)^2 \cdot \\ &= \left(\sum_{n=1}^{\infty} (-1)^{2n} \cdot \frac{(2n-1)!!}{2^n \cdot n!} \cdot x^{2n} \right) + 1 \\ &= \left(\sum_{n=1}^{\infty} (-1)^{2n} \cdot \frac{(2n-1)!!}{(2n)!!} x^{2n} \right) + 1 \end{aligned}$$

$$\therefore \arcsin x = x + \sum_{n=1}^{\infty} (-1)^{2n} \frac{(2n-1)!!}{(2n)!!(2n+1)} x^{2n+1}$$

$$\arccos x = \frac{\pi}{2} - x - \sum_{n=1}^{\infty} (-1)^{2n} \frac{(2n-1)!!}{(2n)!!(2n+1)} x^{2n+1}$$

(8) $\frac{3x-5}{(x-1)(x-3)} = \frac{A}{x-1} + \frac{B}{x-3} = \frac{(A+B)x - 3A - B}{(x-1)(x-3)}$

$$= \frac{1}{x-1} + \frac{2}{x-3} = - \left[(1-x)^{-1} + \frac{2}{3} \cdot \left(1 - \frac{x}{3}\right)^{-1} \right]$$

$$= - \sum_{n=0}^{\infty} x^n - \frac{2}{3} \sum_{n=0}^{\infty} \left(\frac{x}{3}\right)^n$$

$$= - \sum_{n=0}^{\infty} \left(1 + \frac{2}{3^{n+1}}\right) x^n \quad \sqrt[n]{a_n} = \frac{\sqrt[n]{2+3^{n+1}}}{\sqrt[n]{3} \cdot 3} \rightarrow 1$$

$x = \pm 1$, 一般项 $\rightarrow 0$, 不收敛

$$\therefore x \in (-1, 1)$$

$$(7) \arccos x;$$

$$(8) \frac{3x-5}{x^2-4x+3};$$

$$(9) \frac{1}{\sqrt{4-x^2}};$$

$$(10) (1+x)e^{-x};$$

$$(11) \int_0^x \frac{\ln(1+x)}{x} dx;$$

$$(12) \int_0^x \frac{dx}{\sqrt{1-x^4}}.$$

11.4.2 求下列函数在指定点的泰勒展开式及收敛区间

$$\begin{aligned} (9) \quad \frac{1}{2} \cdot \frac{1}{\sqrt{1-\frac{x^2}{4}}} &= \frac{1}{2} \cdot (1-\frac{x^2}{4})^{-\frac{1}{2}} = \frac{1}{2} \cdot (1 + (-\frac{1}{2})(-\frac{x^2}{4}) + \frac{1}{2!}(-\frac{1}{2})(-\frac{3}{2})(-\frac{x^2}{4})^2 + \dots \\ &= \frac{1}{2} \cdot (1 + \sum_{n=1}^{\infty} (-1)^{2n} \frac{(2n-1)!!}{n! \cdot 2^n \cdot 4^n} x^{2n}) \\ &= \frac{1}{2} \cdot (1 + \sum_{n=1}^{\infty} \frac{(2n-1)!!}{2^{2n+1} \cdot n!} x^{2n}) \\ &= \frac{1}{2} + \sum_{n=1}^{\infty} \frac{(2n-1)!!}{(2n)!!} \frac{x^{2n}}{2^{2n+1}} \quad -2 < x < 2. \end{aligned}$$

$$\begin{aligned} (10) \quad \sum_{n=0}^{\infty} \frac{x^n}{n!} \cdot (-1)^n + \sum_{n=0}^{\infty} \frac{x^{n+1}}{n!} \cdot (-1)^n \\ = 1 + \sum_{n=1}^{\infty} (-1)^n \frac{x^n}{n!} + \frac{x^n}{(n-1)!} \cdot (-1)^{n-1}. \end{aligned}$$

$$= 1 + \sum_{n=1}^{\infty} \left(\frac{(-1)^n}{n!} + (-1)^{n-1} \frac{1}{(n-1)!} \right) \cdot x^n$$

$$= 1 + \sum_{n=1}^{\infty} \left[\frac{(-1)^n + (-1)^{n-1} \cdot n}{n!} \right] \cdot x^n$$

$$= 1 + \sum_{n=1}^{\infty} (-1)^{n-1} \cdot \frac{n-1}{n!} x^n, x \in \mathcal{R}$$

$$\begin{aligned} (11) \quad f'(x) &= \frac{\ln(1+x)}{x} = \frac{1}{x} \cdot \sum_{n=1}^{\infty} (-1)^{n-1} \frac{x^n}{n} \\ &= \sum_{n=1}^{\infty} (-1)^{n-1} \frac{x^{n-1}}{n}. \end{aligned}$$

$$\therefore f(x) = \sum_{n=1}^{\infty} (-1)^{n-1} \frac{x^n}{n^2} \quad -1 \leq x \leq 1$$

$$(2) \quad f'(x) = \frac{1}{\sqrt{1-x^4}} = (1-x^4)^{-\frac{1}{2}} = 1 + (-\frac{1}{2})(-x^4) + \frac{1}{2!}(-\frac{1}{2})(-\frac{3}{2})(-x^4)^2 + \dots$$

$$= 1 + \sum_{n=1}^{\infty} (-1)^{2n} \frac{(2n-1)!!}{n! \cdot 2^n} x^{4n}$$

$$= 1 + \sum_{n=1}^{\infty} \frac{(2n-1)!!}{(2n)!!} x^{4n}$$

$$\therefore f(x) = 1 + \sum_{n=1}^{\infty} \frac{(2n-1)!!}{(2n)!!(4n+1)} x^{4n+1}, \quad (-1 < x < 1)$$

11.4.2 求下列函数在指定点的幂级数展开式及其收敛区间:

(1) $\ln x$, 在 $x=1$ 处; (2) $\frac{1}{x^2+3x+2}$, 在 $x=-4$ 处;

(3) $\sqrt{x}$, 在 $x=4$ 处.

11.4.3 求 $f^{(n)}(0)$ ($n=1, 2, 3, \dots$):

(1) $f(x) = \int_0^x \frac{\sin t}{t} dt$; (2) $f(x) = \frac{d}{dx} g(x)$, 其中 $g(x) = \begin{cases} \frac{e^x - 1}{x}, & x \neq 0, \\ 1, & x = 0. \end{cases}$

11.4.4 求下列级数的和:

(1) $\sum_{n=0}^{\infty} \frac{n+1}{n!}$; (2) $\sum_{n=0}^{\infty} \frac{1+n^2}{n!} 2^n x^n$.

$$(1) \text{ 令 } x-1=t, \ln(t+1) = \sum_{n=1}^{\infty} (-1)^{n-1} \frac{t^n}{n}, -1 < t \leq 1, \therefore \ln x = \sum_{n=1}^{\infty} (-1)^{n-1} \frac{(x-1)^n}{n}, 0 < t \leq 2.$$

$$(2) \quad x+4=t, \quad \frac{1}{t^2-8t+16+3t-12+2} = \frac{1}{t^2-5t+6} = \frac{1}{t-3} - \frac{1}{t-2} = \frac{1}{2} \frac{1}{1-\frac{1}{2}t} - \frac{1}{3} \frac{1}{1-\frac{1}{3}t} = \frac{1}{2} \sum_{n=0}^{\infty} \left(\frac{1}{2}t\right)^n - \frac{1}{3} \sum_{n=0}^{\infty} \left(\frac{1}{3}t\right)^n$$

$$= \sum_{n=0}^{\infty} \left(\frac{1}{2^{n+1}} - \frac{1}{3^{n+1}}\right) \cdot t^n, -2 < t < 2$$

$$\therefore \text{原式} = \sum_{n=0}^{\infty} \left(\frac{1}{2^{n+1}} - \frac{1}{3^{n+1}}\right) \cdot (x+4)^n, -6 < x < -2$$

$$(3) \quad t=x-4, \quad \sqrt{4+t} = 2 \sqrt{1+\frac{t}{4}} = 2 \cdot \left(1 + \frac{1}{2} \cdot \frac{t}{4} + \frac{1}{2!} \left(\frac{1}{2}\right) \left(-\frac{1}{2}\right) \cdot \left(\frac{t}{4}\right)^2 + \dots\right)$$

$$= 2 \cdot \left[\sum_{n=2}^{\infty} \frac{(-1)^{n-1}}{2^n} \cdot \frac{(2n-3)!!}{n!} \left(\frac{t}{4}\right)^n \right] + 1 + \frac{t}{8}$$

$$= 2 + \frac{t}{4} + \sum_{n=2}^{\infty} \frac{(-1)^{n-1} (2n-3)!!}{(2n)!! 2^{2n-1}} t^n$$

$$\therefore \sqrt{x} = 2 + \frac{x-4}{4} + \sum_{n=2}^{\infty} \frac{(-1)^{n-1} (2n-3)!!}{(2n)!! 2^{2n-1}} \cdot (x-4)^n$$

11.4.3 求 $f^{(n)}(0) (n=1,2,3,\dots)$:

(1) $f(x) = \int_0^x \frac{\sin t}{t} dt$; (2) $f(x) = \frac{d}{dx} g(x)$, 其中 $g(x) = \begin{cases} \frac{e^x - 1}{x}, & x \neq 0, \\ 1, & x = 0. \end{cases}$

$$(1) f'(x) = \frac{\sin x}{x} = \frac{1}{x} \cdot \sum_{n=1}^{\infty} (-1)^{n-1} \frac{x^{2n-1}}{(2n-1)!} = \sum_{n=1}^{\infty} (-1)^{n-1} \frac{x^{2n-2}}{(2n-1)!}$$

$$f(x) = \sum_{n=1}^{\infty} (-1)^{n-1} \frac{x^{2n-1}}{(2n-1) \cdot (2n-1)!}$$

$$a_{2n} = 0$$

$$a_{2n-1} = (-1)^{n-1} \frac{1}{(2n-1)(2n-1)!} = \frac{f^{(2n-1)}(0)}{(2n-1)!}$$

$$\therefore \begin{cases} f^{(2n)}(0) = 0 \\ f^{(2n-1)}(0) = (-1)^{n-1} \frac{(2n-2)!}{(2n-1)!} = (-1)^{n-1} \frac{1}{2n-1} \end{cases}$$

$$(2) \forall x \neq 0, f(x) = \frac{e^x x - e^x + 1}{x^2} = \frac{(x-1)e^x + 1}{x^2}$$

$$\begin{aligned} g'(0) &= \lim_{\Delta x \rightarrow 0} \frac{g(x+\Delta x) - g(x)}{\Delta x} = \lim_{\Delta x \rightarrow 0} \frac{\frac{e^{x+\Delta x} - 1}{x+\Delta x} - \frac{e^x - 1}{x}}{\Delta x} = \lim_{x \rightarrow 0} \frac{e^x - 1 - x}{x^2} \\ &= \lim_{x \rightarrow 0} \frac{e^x - 1}{2x} \\ &= \frac{1}{2} \end{aligned}$$

$$\therefore f(x) = \begin{cases} \frac{(x-1)e^x + 1}{x^2}, & x \neq 0 \\ \frac{1}{2}, & x = 0 \end{cases}$$

$$x \neq 0 \text{ 时, } g(x) = \frac{1}{x} \cdot \sum_{n=1}^{\infty} \frac{x^n}{n!} = \sum_{n=0}^{\infty} \frac{x^n}{(n+1)!}$$

11.4.4 求下列级数的和:

$$(1) \sum_{n=0}^{\infty} \frac{n+1}{n!}; \quad (2) \sum_{n=0}^{\infty} \frac{1+n^2}{n!} 2^n x^n.$$

11.4.5 求 $\int_0^1 e^{-x^2} dx$ 的近似值, 精确度达到 0.001.

$$(1) \sum_{n=0}^{\infty} \frac{n+1}{n!} = \sum_{n=0}^{\infty} \frac{1}{(n-1)!} + \sum_{n=0}^{\infty} \frac{1}{n!}$$

$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!}, \quad \therefore \sum_{n=0}^{\infty} \frac{1}{n!} = e$$
$$\sum_{n=1}^{\infty} \frac{1}{(n-1)!} = \sum_{n=0}^{\infty} \frac{1}{n!} = e.$$

$$\therefore \text{原式} = 2e$$

$$(2) \sum_{n=0}^{\infty} \frac{1+n^2}{n!} \left(\frac{x}{2}\right)^n = \sum_{n=0}^{\infty} \frac{1}{n!} \left(\frac{x}{2}\right)^n + \sum_{n=1}^{\infty} \frac{n}{(n-1)!} \left(\frac{x}{2}\right)^n$$

$$= e^{\frac{x}{2}} + \sum_{n=1}^{\infty} \frac{1}{(n-2)!} \left(\frac{x}{2}\right)^n + \sum_{n=1}^{\infty} \frac{1}{(n-1)!} \left(\frac{x}{2}\right)^n$$

$$= e^{\frac{x}{2}} + \sum_{n=0}^{\infty} \frac{1}{n!} \left(\frac{x}{2}\right)^{n+2} + \sum_{n=0}^{\infty} \frac{1}{n!} \left(\frac{x}{2}\right)^{n+1}$$

$$= e^{\frac{x}{2}} + \frac{x^2}{4} e^{\frac{x}{2}} + \frac{x}{2} e^{\frac{x}{2}}$$

$$= \left(1 + \frac{x^2}{4} + \frac{x}{2}\right) e^{\frac{x}{2}}$$